

Question

Consider the chemical equation for this reaction: NaBH_4 reacts with nickel chloride (molar ratio 2:1) in an aqueous solution to yield two boron-containing compounds: Ni_xB_y and boric acid (molar ratio 1:3).

Based on the calculated chemical equation for the reaction (all species are electrically neutral, with coefficients as the smallest integers), the following propositions are given:

1. Let the number of reactants be n and the number of products be m ; then $n^m > m^n$.
2. Let the mass fraction of the element with the lowest mass fraction in the species with the highest stoichiometric coefficient among the reactants be ω_x ; let the mass fraction of the element with the lowest mass fraction in the species with the lowest stoichiometric coefficient among the products be ω_y ; then $|\omega_x - \omega_y| < 3\%$.
3. Let the product of the stoichiometric coefficients of all species in the reactants be a , and the product of the stoichiometric coefficients of all species in the products be b ; then $a < b$.
4. Among the reactants in this reaction, there are 3 elements that are reduced.
5. In the reducing agent of this reaction, the element with the highest oxidation state has two different oxidation states in the products.

Select the number of correct propositions from the above.

A. 1, 2, 3, 4, 5

B. 2, 3, 4, 5

C. 1,3,4,5

D. 1, 2, 4, 5

E. 1, 2, 3, 5

F. 1, 2, 3, 4

G. 1, 2, 3

H. 1, 2, 4

I. 2,3,4

J. 3,4,5

K. 2, 3, 5

L. None of the above options are accurate

Answer

Correct Answer: **A**

Detailed Explanation

The problem describes the reaction of sodium borohydride (NaBH_4) with nickel chloride (NiCl_2) in aqueous solution.

First, it is necessary to determine the reactants and products of the reaction. Since boric acid is a reaction product, water must participate in the reaction, and the reactants are NaBH_4 , NiCl_2 , H_2O (aqueous solution environment).

CHECKPOINT

1 PTS

Reactants are NaBH_4 , NiCl_2 , H_2O

Since NaBH_4 reduces other species, its own negative hydrogen is oxidized to form hydrogen gas (H_2), and Cl^- in the reactants exists in the product as sodium chloride (NaCl), so the reaction products are Ni_xB_y , H_3BO_3 , H_2 , NaCl .

CHECKPOINT

1 PTS

Reaction products are Ni_xB_y , H_3BO_3 , H_2 , NaCl

The chemical reaction equation containing unknowns can be obtained as:



According to the key stoichiometric ratios given in the problem:

1. The molar ratio of reactants $\text{NaBH}_4 : \text{NiCl}_2$ is $2 : 1$.
2. The molar ratio of products $\text{Ni}_x\text{B}_y : \text{H}_3\text{BO}_3$ is $1 : 3$.

The equation is simplified to:



The following element conservation applies:

- Ni element conservation: $b = xd$
- B element conservation: $2b = (y + 3)d$
- Na/Cl element conservation: $2b = f$
- H element conservation: $8b + 2c = 9d + g$
- O element conservation: $c = 9d$

From the element conservation of Ni and B, $2x = y + 3$ in Ni_xB_y . It is impossible to determine the only possibility. Assume that $x = 2, y = 1$ is the most likely answer ($x = 3, y = 3$ is unreasonable). Therefore, Ni_xB_y is most likely Ni_2B .

CHECKPOINT

1 PTS

 Ni_xB_y is Ni_2B

Combining the above information, the final chemical equation is:



CHECKPOINT

1 PTS

The final chemical equation is:



Step 1: Determine the chemical reaction equation

The problem describes the reaction of sodium borohydride (NaBH_4) with nickel chloride (NiCl_2) in aqueous solution. This is a chemical reaction used to prepare amorphous nickel boride catalysts.

Reactants: NaBH_4 , NiCl_2 , H_2O (aqueous solution environment).

Products: Nickel boride (Ni_xB_y , commonly Ni_2B), boric acid (H_3BO_3), sodium chloride (NaCl), hydrogen (H_2).

Further analyze the five propositions one by one

Proposition 1: Let the number of reactants be n , and the number of products be m , then $n^m > m^n$.

Reactants: NaBH_4 , NiCl_2 , H_2O . So $n = 3$.

Products: Ni_2B , H_3BO_3 , NaCl , H_2 . So $m = 4$.

It is necessary to compare n^m and m^n , that is, compare 3^4 and 4^3 . Since $81 > 64$, $n^m > m^n$ holds. Therefore, Proposition 1 is correct.

CHECKPOINT

1 PTS

Proposition 1 is correct

Proposition 2: Let ω_x be the mass fraction of the element with the lowest mass fraction in the species with the highest stoichiometric number among the reactants; let ω_y be the mass fraction of the element with the lowest mass fraction in the species with the lowest stoichiometric number among the products, then $|\omega_x - \omega_y| < 3\%$.

Reactant coefficients: 8, 4, 18. The species with the highest stoichiometric number is H_2O (coefficient is 18).

The molar mass of H_2O is $M \approx 18.016 \text{ g/mol}$.

The element with the lowest mass fraction is hydrogen (H).

$$\omega_x = \omega(\text{H, in H}_2\text{O}) = \frac{2 \times 1.008}{18.016} \approx 0.1119 = 11.19\%$$

Product coefficients: 2, 6, 8, 25. The species with the lowest stoichiometric number is Ni_2B (coefficient is 2).

The molar mass of Ni_2B is $M = (2 \times 58.69) + 10.81 = 128.19 \text{ g/mol}$.

The element with the lowest mass fraction is boron (B).

$$\omega_y = \omega(\text{B, in Ni}_2\text{B}) = \frac{10.81}{128.19} \approx 0.0843 = 8.43\%$$

Comparison:

$$|\omega_x - \omega_y| = |11.19\% - 8.43\%| = 2.76\%$$

Since $2.76\% < 3\%$, $|\omega_x - \omega_y| < 3\%$ holds. Therefore, Proposition 2 is correct.

CHECKPOINT

1 PTS

$$|\omega_x - \omega_y| = |11.19\% - 8.43\%| = 2.76\%, \text{ Proposition 2 is correct}$$

Proposition 3: Let a be the product of the stoichiometric numbers of all species in the reactants, and b be the product of the stoichiometric numbers of all species in the products, then $a < b$.

Reactant stoichiometric numbers: 8, 4, 18.

$$a = 8 \times 4 \times 18 = 576$$

Product stoichiometric numbers: 2, 6, 8, 25.

$$b = 2 \times 6 \times 8 \times 25 = 2400$$

Since $576 < 2400$, $a < b$ holds. Therefore, Proposition 3 is correct.

CHECKPOINT

1 PTS

$a < b$, Proposition 3 is correct

Proposition 4: In the reactants of this reaction, 3 elements are reduced.

Analyze the changes in the valence of each element before and after the reaction:

- * Ni: +2 in NiCl_2 , valence decreases in Ni_2B . Reduced.
- * H: +1 in H_2O , partially converted to H_2 (0 valence). Reduced.
- * B: +3 in NaBH_4 , partially converted to Ni_2B (negative valence). Reduced.
- * H (another form): -1 in NaBH_4 , converted to H_2 (0 valence). Oxidized.
- * Na, Cl, O: The valence does not change before and after the reaction.

Therefore, the elements that are reduced are Ni, H, B. Therefore, Proposition 4 is correct.

CHECKPOINT

1 PTS

The elements that are reduced are Ni, H, B, Proposition 4 is correct

Proposition 5: In the reducing agent of this reaction, the element with the highest valence has two different valences in the products.

Reducing agent: NaBH_4 .

Element valences in the reducing agent: $\text{Na}(+1)$, $\text{B}(+3)$, $\text{H}(-1)$.

Element with the highest valence: $\text{B}(+3)$.

Valence of this element (B) in the products:

1. In H_3BO_3 , the valence of B is +3.
2. In Ni_2B , the valence of B is negative.

Therefore, the element B with the highest valence does have two different valences in the products. Therefore, Proposition 5 is correct.

CHECKPOINT

1 PTS

In H_3BO_3 , the valence of B is +3, while in Ni_2B , the valence of B is negative, so Proposition 5 is correct

After analysis, Propositions 1, 2, 3, 4, and 5 are all correct. The final answer selects the corresponding option **A**.